

RESEARCH PROFILE

My research studies the **structure, computation, and learning of values, equilibria, and strategies in stochastic games**, in which the same players interact again and again and present choices shape future interaction. During my Ph.D., I have studied four classes of such games: concurrent stochastic games, turn-based stochastic games, partially observable Markov decision processes, and robust Markov decision processes. I investigate these games from three perspectives:

- **Mathematical structure:** existence and characterization of values, equilibria, and optimal strategies, and the algebraic structure, memory, and numerical precision they require.
- **Computation:** the complexity of these problems, efficient algorithms, and the guarantees and limits of classical methods such as policy and strategy iteration.
- **Learning:** what can be learned about values and strategies from finitely many observations, and how robust models account for estimation error.

EDUCATION

- **Ph.D. in Computer Science** 2021 – 2027 (expected)
Institute of Science and Technology Austria (ISTA), Klosterneuburg, Austria
Advisor: [Krishnendu Chatterjee](#)
- **B.Sc. in Computer Engineering** 2016 – 2021
Sharif University of Technology, Tehran, Iran
GPA: 18.93/20

PUBLICATIONS

- “Algorithms with Smoothed Polynomial-Time Complexity for Deterministic Discounted-sum and Mean-payoff Games”
Ali Asadi, Krishnendu Chatterjee, Ruichen Luo
to appear in *the 38th Annual ACM-SIAM Symposium on Discrete Algorithms (SODA’27)*, 2027.
- “On the Complexity of Discounted Robust MDPs with L_p Uncertainty Sets”
Ali Asadi, Krishnendu Chatterjee, Alipasha Montaseri, Ali Shafiee
to appear in *the 40th Annual Conference on Neural Information Processing Systems (NeurIPS’26, Spotlight)*, 2026.
- “Generalized Bidding Games: Where Bidding and Stochastic Games Meet”
Ali Asadi, Thomas A. Henzinger, Ehsan Kafshdar Goharshady, Pavol Kebis, Kaushik Mallik
in *the 37th International Conference on Concurrency Theory (CONCUR’26)*, 2026.
- “PAC Learning in Turn-Based Stochastic Games with Reachability Objectives: A Decentralized Private Approach via Expected Conditional Distance”
Ali Asadi, Krishnendu Chatterjee, Pavol Kebis
in *the 37th International Conference on Concurrency Theory (CONCUR’26)*, 2026.
- “Strongly Polynomial Time Complexity of Policy Iteration for L_∞ Robust MDPs”
Ali Asadi, Krishnendu Chatterjee, Ehsan Goharshady, Mehrdad Karrabi, Alipasha Montaseri, Carlo Pagano
in *the 39th Annual Conference on Learning Theory (COLT’26)*, 2026.
- “Qualitative Analysis of ω -Regular Objectives on Robust MDPs”
Ali Asadi, Krishnendu Chatterjee, Ehsan Goharshady, Mehrdad Karrabi, Ali Shafiee
in *the 40th Annual AAAI Conference on Artificial Intelligence (AAAI’26)*, 2026.
- “Revealing POMDPs: Qualitative and Quantitative Analysis for Parity Objectives”
Ali Asadi, Krishnendu Chatterjee, David Lurie, Raimundo Saona
in *the 40th Annual AAAI Conference on Artificial Intelligence (AAAI’26)*, 2026.

- “ [\$\epsilon\$ -Stationary Nash Equilibria in Multi-player Stochastic Graph Games](#)”
Ali Asadi, Léonard Brice, Krishnendu Chatterjee, K. S. Thejaswini
in the 45th IARCS Annual Conference on Foundations of Software Technology and Theoretical Computer Science (**FSTTCS’25**), 2025.
- “[Limit-sure reachability for small memory policies in POMDPs is NP-complete](#)”
Ali Asadi, Krishnendu Chatterjee, Raimundo Saona, Ali Shafiee
in the 41st Conference on Uncertainty in Artificial Intelligence (**UAI’25**), 2025.
- “[Lower Bound on Howard Policy Iteration for Deterministic Markov Decision Processes](#)”
Ali Asadi, Krishnendu Chatterjee, Jakob de Raaij
in the 41st Conference on Uncertainty in Artificial Intelligence (**UAI’25**), 2025.
- “[Concurrent Stochastic Games with Stateful-discounted and Parity Objectives: Complexity and Algorithms](#)”
Ali Asadi, Krishnendu Chatterjee, Raimundo Saona, Jakub Svoboda
in the 44th IARCS Annual Conference on Foundations of Software Technology and Theoretical Computer Science (**FSTTCS’24**), 2024.
- “[Deterministic Sub-exponential Algorithm for Discounted-sum Games with Unary Weights](#)”
Ali Asadi, Krishnendu Chatterjee, Raimundo Saona, Jakub Svoboda
in the 39th Annual Symposium on Logic in Computer Science (**LICS’24**), 2024.
- “[Polynomial reachability witnesses via Stellensätze](#)”
Ali Asadi, Krishnendu Chatterjee, Hongfei Fu, Amir Kafshdar Goharshady, Mohammad Mahdavi
in the 42nd ACM SIGPLAN International Conference on Programming Language Design and Implementation (**PLDI’21**), 2021.
- “[Faster Algorithms for Quantitative Analysis of MCs and MDPs with Small Treewidth](#)”
Ali Asadi, Krishnendu Chatterjee, Amir Kafshdar Goharshady, Kiarash Mohammadi, Andreas Pavlogiannis
in the 18th International Symposium on Automated Technology for Verification and Analysis (**ATVA’20**), 2020.

MANUSCRIPTS UNDER REVIEW

- “[Quantitative Analysis of \$\omega\$ -Regular Robust MDPs](#)”
Ali Asadi, Krishnendu Chatterjee, Ehsan Kafshdar Goharshady, Mehrdad Karrabi, Alipasha Montaseri, Ali Shafiee
submitted to the 41st Annual AAAI Conference on Artificial Intelligence (**AAAI’27**).
- “[The Complexity of Approximating the Value in Revealing POMDPs with Long-Run Average Objectives](#)”
Ali Asadi, Krishnendu Chatterjee, David Lurie
submitted to *Mathematics of Operations Research* (**MOOR**).
- “[Values of Absorbing Recursive Games Express All Real Algebraic Numbers](#)”
Ali Asadi, Krishnendu Chatterjee
submitted to *Mathematics of Operations Research* (**MOOR**).

TALKS AND POSTERS

- “[Revealing POMDPs: Qualitative and Quantitative Analysis for Parity Objectives](#)”
Talk: Highlights of Logic, Games and Automata (Highlights’26), Vienna, Austria Sept. 2026
Poster: Paris Game Theory Festival 2026, Paris, France June 2026
Poster: AAAI Conference on Artificial Intelligence (AAAI’26), Singapore Jan. 2026
Seminar talk: Combinatorics, Games and Optimisation Seminar, LSE, London, UK Dec. 2025
- “[Limit-sure reachability for small memory policies in POMDPs is NP-complete](#)”
Poster: 30 Years of Game Theory at Institut Henri Poincaré, Paris, France Oct. 2025

- “Deterministic Sub-exponential Algorithm for Discounted-sum Games with Unary Weights”
Talk: International Conference on Reachability Problems (RP’24), Vienna, Austria Sept. 2024
Talk: ACM/IEEE Symposium on Logic in Computer Science (LICS’24), Tallinn, Estonia July 2024
Poster: Mediterranean Game Theory Symposium 2024, Marseille, France May 2024

HONORS AND AWARDS

- **Silver medal** in the 10th Asia-Pacific Informatics Olympiad (APIO) 2016
- **Gold and Silver medals** in the 25th and 24th Iranian National Olympiad in Informatics (INOI) 2015, 2014
- **2nd place** in the ACM-ICPC West Asia Regional Contest, Tehran, Iran. 2016
- Iranian National Elites Foundation grant for undergraduate studies 2014 – 2021
- Research Scholarship from the Austrian Exchange Agency (OeAD-GmbH) 2019

SCIENTIFIC SERVICE

- **Program Committee Member**
 - AAAI Conference on Artificial Intelligence (AAAI’27) 2027
 - AAAI Conference on Artificial Intelligence (AAAI’26) 2026
- **Journal Reviewer**
 - SIAM Journal on Control and Optimization (SICON) 2026
 - Information and Computation 2026
- **Conference Reviewer**
 - Conference on Uncertainty in Artificial Intelligence (UAI’26) 2026
 - Foundations of Software Technology and Theoretical Computer Science (FSTTCS’25) 2025
 - International Symposium on Mathematical Foundations of Computer Science (MFCS’25) 2025
 - International Colloquium on Automata, Languages, and Programming (ICALP’24) 2024

TEACHING EXPERIENCE

University Teaching

- **Teaching Assistant, Institute of Science and Technology Austria (ISTA)** 2024 – 2025
Held office hours and prepared and graded homework assignments.
 - Computer Science Track Core Course Spring 2024; Spring 2025
 - Formal Methods: Algorithmic Approaches Spring 2024
- **Teaching Assistant, Sharif University of Technology** 2017 – 2018
Prepared and graded homework assignments.
 - Data Structures and Algorithms Spring 2018
 - Probability and Statistics Fall 2017

Olympiad Teaching and Educational Service

- **Lecturer, Algorithms and Data Structures** 2016 – 2019
Prepared students for the Iranian National Olympiad in Informatics (INOI).
 - Allameh Helli High School 2016 – 2018
 - Allameh Tabatabaee High School 2016 – 2019
- **Scientific Committee Member and Problem Author, INOI**
 - INOI Final Exams 2022 – 2025
 - IOI Team Selection Exams 2017
 - INOI Summer Camp Programming Exams 2016

RESEARCH SUPERVISION

Research Interns and Rotation Students, Institute of Science and Technology Austria (ISTA)

- **Pavol Kebis** Fall 2025
 - “PAC Learning in Turn-Based Stochastic Games with Reachability Objectives: A Decentralized Private Approach via Expected Conditional Distance” (**CONCUR’26**)
- **Jack Quirion** Summer 2025
 - “Multi-Objective Reachability in Stochastic Games” (**planned submission to ICALP’27**)
- **Alipasha Montaseri** Summer 2025
 - “Strongly Polynomial Time Complexity of Policy Iteration for L_∞ Robust MDPs” (**COLT’26**)
 - “On the Complexity of Discounted Robust MDPs with L_p Uncertainty Sets” (**NeurIPS’26, Spotlight**)
- **Ali Shafiee** Summer 2024
 - “Qualitative Analysis of ω -Regular Objectives on Robust MDPs” (**AAAI’26**)
 - “Limit-sure reachability for small memory policies in POMDPs is NP-complete” (**UAI’25**)
- **Jakob de Raaij** Summer 2023
 - “Lower Bound on Howard Policy Iteration for Deterministic Markov Decision Processes” (**UAI’25**)

INDUSTRY EXPERIENCE

- **Engineering Manager, Divar Co.** Sept. 2019 – Sept. 2021
Led a team of 10 engineers building scalable advertising infrastructure.
- **Software Engineer, CafeBazaar Co.** July 2017 – July 2019
Developed large-scale payment and deployment systems.